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SHORT COMMUNICATION

First detection of tilapia lake virus (TiLV) in wild river carp (*Barbonymus schwanenfeldii*) at Timah Tasoh Lake, Malaysia

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Tilapia lake virus (TiLV) is a new emerging disease in tilapia announced by the NACA & OIE organizations since early 2017 (Dong, Siriroot, et al., 2017; OIE, 2013). The mortality of tilapines due to this virus was observed as early as 2009 in Israel, however without any confirmation of the causative agent(s). In 2014, they have confirmed that this mortality was caused by a virus from family *orthomyxoviridae* and temporarily designated as TiLV (Eyngor et al., 2014). At present, TiLV has been reported in several Asian countries. Several publications and notifications to the OIE have been made by the Chinese Taipei (OIE, 2017b), India (Behera et al., 2018), Israel (OIE, 2017e), Thailand (OIE, 2017d), Malaysia (Amal et al., 2018; OIE, 2017c) and the Philippines (OIE, 2017a). Other than the above countries, Ecuador, Colombia and Egypt have also been reported to be affected by TiLV (Fathi et al., 2017; Ferguson et al., 2014; Tsoufack et al., 2017). So far, TiLV has only been reported to cause disease in farmed and wild tilapia (Eyngor et al., 2014).

In June 2017, a mass mortality involving wild fish in a man-made lake at northern part of Malaysia was reported. Wild tilapia was the most affected fish species followed by barbs, which includes river barb (*Barbonymus schwanenfeldii*) and java barb (*Barbonymus gonionotus*). Other fish species such as catfish and peacock bass (*Cichla* sp.) were also affected but at a very low rate. The mass mortality incidence occurred following the release of water for irrigation, thus reduced the water level of the lake. The long and hot dry season was also noted at this area, which further reduced the water level of the lake. It was reported that about 300–500 fish had died. The main species affected was wild tilapia of 500–600 g, followed by river barb, a small number of peacock bass and freshwater catfish. During investigation, the mortality has been reduced and only involved tilapia, with some dead and moribund fish found floating on the water surface. The size of the moribund fish ranged from 200 to 600 g. Most of the dead fish were found at the water inlet gates.

A total of 10 black tilapias were caught and observed for external clinical signs. Four were dissected on-site. In addition, five healthy river barbs were also sampled and dissected on-site. Physical examination on tilapia showed reddish discoloration of the body, operculum, ventral and lateral fins (Figure 1a). The eyes appeared normal. Two of four fish dissected showed a watery, soft or haemorrhagic brain. River barb were physically healthy, and the organs were also normal, except mild congested kidney and pale liver in one fish (Table 1). For bacteriological examination, samples were aseptically taken from brain, eye and kidney using sterile loop and streaked onto tryptic soy agar (Oxoid, Hampshire, UK). Organ samples for PCR were stored in 70% ethanol. Sampling of water quality at the surface and 4 m depth was carried out at three main points, inlet, centre and outlet of the lake, and analysed for physical and chemical parameters. All fish and water samples were brought back to National Fish Health Research Centre (NaFisH) for further analysis.

Bacteriology plates were incubated at 30°C for 24–48 hr. Biochemical analysis was performed using API 20E, API 20NE and API 20STAPH (Biomérieux, France). Total nucleic acid was extracted using viral DNA/RNA gene spin kit (iNtRON Biotechnology, Gyeonggi-do, South Korea). The extracted products were subjected to semi-nested reverse transcriptase–polymerase chain reaction technique (RT-PCR) suggested by Dong, Rattanarojpong, and Senapin (2017). Positive control for TiLV was obtained from Dr. Thanh Ha Dong, Centex Shrimp, Mahidol University, Thailand. Purified PCR products were sent for sequencing. The evolutionary history and analysis were inferred using the neighbour-joining method (Saitou & Nei, 1987) conducted in MEGA7 (Kumar, Stecher, & Tamura, 2016). The samples were also tested for VNN and iridovirus as previously described (Abdullah et al., 2017).

The results of PCR showed that all tilapia samples were negative for VNN and iridovirus. However, the results were positive for TiLV

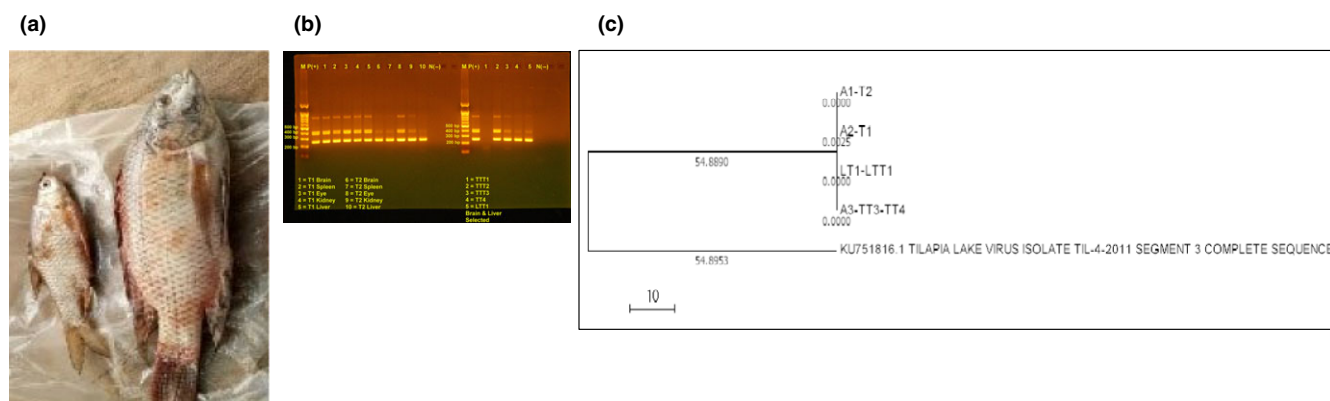


FIGURE 1 (a) Clinical signs observed in river carp (left) and wild tilapia (right) showing reddish discoloration of fins, body and scales. Photo courtesy: Perlis Biosecurity Division, Department of Fisheries Malaysia. (b) Gel electrophoresis–polymerase chain reaction (PCR) technique showed infected tilapia (TT) and river carps (LTT). N = negative control; M = DNA marker; P = positive control; 1 = TT1; 2 = TT2; 3 = TT3; 4 = TT4; 5 = LTT1. Photo courtesy (b): Lab-Ind. Resources Sdn. Bhd. (c) Phylogenetic tree showing similarity of the sequence from this study with Israel strain

TABLE 1 Summary of tests results conducted on the tilapia and river carps fish during on-site investigation

	Wild Tilapia (<i>Oreochromis niloticus</i>)	River carps (<i>Puntius schwanefeldii</i>)
Total number of live samples	4	5
Size of fish	400–600 g	80–200 g
Physical examination	Reddish discoloration on the body and fins (3/4) especially at peduncle region Rotten gills (1/4) Eye—normal	Apparently healthy and normal
Organ examination	Mild haemorrhage of the brain (2/4) Pale/patchy liver (2/4) Congested kidney (2/4)	Pale liver (1/5) Congested kidney (1/5)
Bacterial isolation	<i>Staphylococcus xylosum</i> (1/4 from fish kidney) <i>Aeromonas sobria</i> (3/4 from fish brain, eye and kidney)	<i>Staphylococcus xylosum</i> (2/5) <i>Aeromonas sobria</i> (1/5) from fish brain, eye & kidney) <i>Plesiomonas shigelloides</i> .
Virology	No of samples: 3 Positive TiLV (3/3) Negative VNN and iridovirus	No of samples: 2 Positive TiLV (2/2) Negative VNN and iridovirus

in all tilapia samples, and surprisingly, the results were also positive in apparently healthy river barb (Figure 1b). The amplicon sent for sequencing showed that sequence from both tilapia and river barb is 96% similar to Segment 3 TiLV isolate Til-4-2011 from Israel with Accession No. KU751816.1 (Figure 1c).

TiLV in river barb or any other freshwater fish species has never been reported (Jansen & Mohan, 2017). Thus, our finding has broadened the expectations that TiLV could infect other freshwater fish even at wild. Previous studies found that when tilapia was positive TiLV, the species being cultured together is TiLV-negative. The species includes grey mullet (*Mugil cephalus*) and barb (*Cyprinus barbilio*) in Israel (Eyngor et al., 2014), grey mullet and thin-lipped mullet (*Liza ramada*) in Egypt (Fathi et al., 2017) and co-cultivated Indian Major Barbs (rohu) (*Labeo rohita*), catla (*Catla Catla*), mrigal (*Cirrhinus mrigala*), milkfish (*Chanos chanos*) and pearl spot (*Etroplus suratensis*) in India (Behera et al., 2018).

Other than TiLV, several bacteria species have been isolated from both tilapia and river barb. Results from API Kit show bacteria resemblance to *Staphylococcus xylosum* and *Aeromonas sobria* isolated from tilapia and river barb, and *Plesiomonas shigelloides* isolated from river barb (Table 1). These bacteria are commonly found in the environment, and its association with the occurrence of TiLV is unknown.

Water quality at inlet gates showed lower dissolved oxygen as compared to the other sampling sites (Table 2). It was reported that in normal condition, the dissolved oxygen in the lake was 9.93–16.48 mg/L (Abdullah, Hasan, Kamal, & Hashim, 2014). The ammonia content was 0.27 mg/L at the outlet and 0.3 mg/L, above recommended level of aquaculture activities. Other parameters were within normal range.

The occurrence of mass mortality in several fish species in Timah Tasoh Lake, mainly of the wild tilapia and river barb, is unexpected.

TABLE 2 Water quality parameters reveal high ammonia contents, above the range suggested for aquaculture (<0.02 ppm). One of the sampling locations has low dissolved oxygen (Bold)

Parameter	Water inlet		Centre		Water outlet	
	Surface	4 m	Surface	4 m	Surface	4 m
Temperature (°C)	31.5	29.0	31.6	30.3	31.7	30.6
Dissolved oxygen (mg/L)	5.58	3.34	7.15	5.06	5.83	5.06
pH	8.09	7.97	8.44	8.29	8.18	8.09
Ammonia (mg/L)	0.30		0.28		0.27	
Ferum (mg/L)	0.15		0.18		0.06	
Sulphide (µg/L)	8		10		9	
Nitrite (mg/L)	0.003		0.002		0.002	

Water quality, weather and water level may influence the occurrence of TiLV. The introduction of new species and recreational activities further increases this possibility. A previous report (Abdullah et al., 2014) showed that prolonged dry season recedes the water level makes it become shallow, thus changing the equilibrium between the sediment and the water. Mortality of other fish species such as river barb, catfish and peacock at the beginning of the incidence may indicate a water quality problem at the base of the lake as these species of fish usually stay at the bottom of the lake. Low oxygen content at deeper area and at water inlet together with secondary bacterial infection may cause stress to these fish and unfortunately enhance the TiLV infection to river barb (Amal et al., 2018).

River barb are among the cyprinid family that have been abundantly found throughout water bodies in Malaysia. Barbs and tilapia have been a major protein source for the poor countries, and the emergence of TiLV in species other than tilapia will have a huge impact on the industries locally and globally. With barbs in the list of susceptible species for TiLV, and such situations are likely to be present in several countries in the region, the freshwater fish industry would be highly at risk in contracting the virus.

While we have detected TiLV in samples from apparently healthy river barb, we do not know whether the virus is potentially infectious and whether the river barb or other species in the similar family may act as subclinical shedders. Furthermore, there is a tendency of one pathogen may enhance the other pathogen, and in our case, we found multiple bacterial isolated from fish organs together with TiLV, even in healthy barbs. This finding, however, needs to be analysed further.

The mortality event in this lake took about 50 days to fully stop. Based on the available information, we suggest that the peak mortality is occurring between eight and 13 days after the onset of mortality. The mortality in tilapia has reduced after a month, but the small quantity of river barb was still dying until one and a half month, suggesting the longer period of virus shedding from this species.

In this investigation, virus culture was not performed. Therefore, it is unknown whether the positive PCR signal in the river barb represented live viral particles. The epidemiological significance of this detection therefore remains unknown. Future studies should aim to determine whether the river barb is a fully susceptible species for

TiLV, and their potential role in the epidemiology and spread of TiLV should be investigated in detail. During the preparation of this manuscript, this is the first reported case of TiLV outbreak occurred in the wild fish in Malaysia that infects not only tilapia but also river barb. Hence, we should consider the possibilities of other freshwater fish especially in barbs family to be infected by this virus, and further, thorough studies must be carried out to prove this theory.

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